

Isaac ROS Object detection

Isaac ROS Object detection

[Overview](#)

[Quick Experience](#)

Isaac ROS Object Detection Official Website Link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_object_detection/index.html

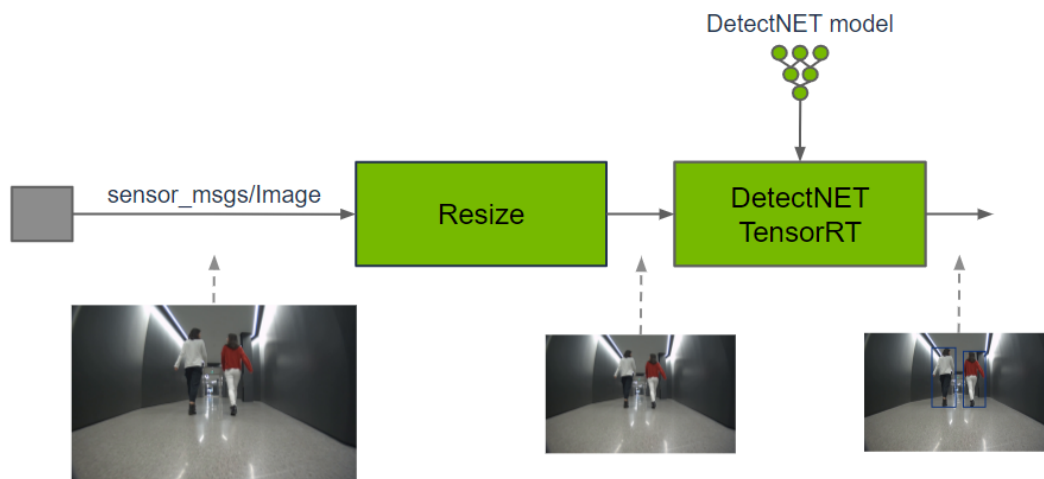
Overview

Isaac ROS Object Detection contains ROS 2 packages to perform object detection.

`isaac_ros_rtdetr`, `isaac_ros_detectnet`, and `isaac_ros_yolov8` each provide a method for spatial classification using bounding boxes with an input image. Classification is performed by a GPU-accelerated model of the appropriate architecture:

- `isaac_ros_rtdetr`: RT-DETR models
- `isaac_ros_detectnet`: DetectNet models
- `isaac_ros_yolov8`: YOLOv8 models

The output prediction can be used by perception functions to understand the presence and spatial location of an object in an image.



Quick Experience

To simplify development, we primarily use the Isaac ROS Dev Docker image and demonstrate the effects there. This demonstration does not require any camera device installation; it simulates the data stream from a camera by playing a rosbag file.

Note: If you wish to install on your own device or connect a camera to develop other features, please refer to the Isaac ROS official website and connect to an NVIDIA-specified camera model for your own development.

Open a terminal and enter the working directory.

```
cd ${ISAAC_ROS_WS}/src
```

Enter the Isaac ROS Dev Docker container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following startup command.

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py  
••launch_fragments:=detectnet  
interface_specs_file:=${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_detectnet/quick  
start_interface_specs.json
```

Open a second terminal and enter the container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 bag play -l  
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_detectnet/rosbags/detectnet_rosbag --  
remap image:=image_rect camera_info:=camera_info_rect
```

View the run results

Open a third terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 run isaac_ros_detectnet isaac_ros_detectnet_visualizer.py --ros-args --remap  
image:=detectnet_encoder/resize/image
```

Open a fourth terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command to view the results.

```
ros2 run rqt_image_view rqt_image_view /detectnet_processed_image
```

